

Estruturas de Dados

Java



DIFERENTES ESTRUTURAS DE DADOS

LIST	Permite elementos duplicados Ordem dos elementos é a ordem de inserção
SET	Não permite duplicatas de objetos Portanto o objeto a ser inserido precisa ter o equals e (desejável) o hashCode implementados
MAP	“Dicionário” de elementos que são identificados por chave/valor Pode haver valores duplicados? Desde que as chaves sejam distintas

APIs ESSENCIAIS DO JAVA

01

OPTIONAL

02

DATE AND TIME

03

REFLECTION

04

ANNOTATION PROCESSING

05

REGEX

06

SECURITY

Java IO

Desde o Java 1.0

- Baseada em streams (fluxos): InputStream, OutputStream, Reader, Writer
- Bloqueante (blocking I/O): uma thread fica esperando a operação concluir.
- Orientada a byte ou caractere.
- Lida com arquivos, buffers de memória e dispositivos de entrada/saída (como teclado e sockets).

Java NIO

A partir do Java 1.4

- Baseada em buffers e canais (Channels), não streams.
- Não bloqueante (non-blocking I/O): uma thread pode continuar enquanto outra espera.
- Permite multiplexação com Selectors, útil para aplicações com múltiplas conexões (como servidores).
- Mais eficiente para leitura e escrita de grandes volumes de dados (como arquivos binários ou redes).

Java NIO.2

A partir do Java 7

- Introduz a nova classe `java.nio.file.Path` substituindo `File`.
- Trabalha com `Files`, `Path`, `FileSystem` e `WatchService`.
- Permite manipulação moderna e robusta de arquivos e diretórios.
- Suporte a sistema de arquivos virtual, como ZIP e sistemas remotos.
- Suporte a monitoramento de arquivos (watching file changes).
- Operações atômicas, cópia, mover, deletar, e muito mais.

Gerenciamento Transacional

Toda transação tem as seguintes propriedades

ACID

A

ATOMICIDADE

C

CONSISTENCIA

I

ISOLAMENTO

D

DURABILIDADE

Toda transação tem as seguintes propriedades

ACID

◀◀

A

ATOMICIDADE

Regra do Tudo ou Nada
A transação é realizada completamente ou
retorna a um estado anterior à sua execução

▶

Toda transação tem as seguintes propriedades

ACID



C

CONSISTENCIA

O banco de dados parte de um estado consistente (antes da transação) e se mantém consistente (após a transação)



Toda transação tem as seguintes propriedades

ACID



ISOLAMENTO

Transações podem ocorrer em paralelo
Sem que uma interfira na outra



Toda transação tem as seguintes propriedades

ACID



D

DURABILIDADE

Uma vez a transação concluída, suas alterações seguem permanentes

COMPANY HISTORY

1XXX

VENUS

Venus is the second planet from the Sun

1XXX

MARS

Mars is full of iron oxide dust

2XXX

JUPITER

Jupiter doesn't have a solid surface

2XXX

SATURN

Saturn was named after a Roman god



COMPANY MILESTONE

MILESTONE 1

Mercury is the closest planet to the Sun

MILESTONE 3

Venus is the farthest planet from the Sun

MILESTONE 5

Mars is actually a very cold place

MILESTONE 2

Jupiter is the biggest planet of them all

MILESTONE 4

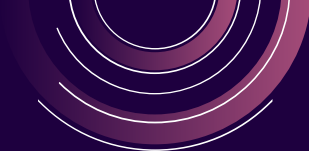
Saturn is composed of hydrogen and helium



<60%

COMPANY'S GROWTH

Mars is actually a cold place



200

PARTNERS

Venus has a beautiful name

4,000

EMPLOYEES

Mercury is a very small planet

450,000

USERS

Saturn has several rings



EMPLOYEE BENEFITS

WORK-LIFE BALANCE

Venus is the second planet from the Sun



HEALTH INSURANCE

Pluto is now considered a dwarf planet



CATERING SERVICE

Saturn is a gas giant and has several rings



RETIREMENT PLANS

Neptune is the farthest planet from the Sun



JOB DESCRIPTION 1

DESIRED PROFILE

- Describe the desired profile here
- Describe the desired profile here
- Describe the desired profile here
- Describe the desired profile here

REQUIRED STUDIES

- Describe the required studies here
- Describe the required studies here
- Describe the required studies here
- Describe the required studies here

REQUIRED SKILLS

- Describe the required skills here
- Describe the required skills here
- Describe the required skills here
- Describe the required skills here



JOB DESCRIPTION 2



ABOUT THE COMPANY

Despite being red, Mars is a cold place

JOB OBJECTIVES

Mercury is the closest planet to the Sun and the smallest one in the Solar System

RESPONSIBILITIES

- Describe the responsibilities for this job position



JOB DESCRIPTION 3

JOB TITLE

Software developer engineer

DEPARTEMENT

Software development

LOCATION

New York

DURATION

12 months

WORK SCHEDULE

40 hours per week / 52 weeks per year

JOB OBJECTIVE

You can briefly describe the job position and the objectives to be pursued

RESPONSIBILITIES

- Describe the responsibilities here
- Describe the responsibilities here
- Describe the responsibilities here

EDUCATION & EXPERIENCE

- Describe the education here
- Describe the experience here

OBJECTIVES & RESPONSIBILITIES OF THE ROLE

JOB OBJECTIVE

- Describe the first job objective here
- Describe the second job objective here
- Describe the third job objective here

RESPONSIBILITIES

- Describe the first responsibility here
- Describe the second responsibility here
- Describe the third responsibility here
- Describe the fourth responsibility here
- Describe the fifth responsibility here
- Describe the sixth responsibility here
- Describe the seventh responsibility here

GOALS

COMMUNICATION GOALS

- Describe the different goals of the role here
- Describe the different goals of the role here
- Describe the different goals of the role here
- Describe the different goals of the role here

MARKETING GOALS

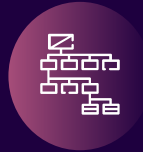
- Describe the different goals of the role here
- Describe the different goals of the role here
- Describe the different goals of the role here
- Describe the different goals of the role here

VALUABLE KNOWLEDGE



PROGRAMMING TOOLS

Despite being red,
Mars is a cold place



TEAMWORK

Saturn is gas giant and
has several rings



LANGUAGE

Mercury is the smallest
planet of them all



LEADERSHIP

Venus is the second
planet from the Sun

REQUIRED KNOWLEDGE

REQUIRED KNOWLEDGE

PROGRAMMING

Venus is a hot planet

JAVA

Jupiter doesn't have
a solid surface

PYTHON

Mars is full of iron
oxide dust

LANGUAGES

Saturn is a gas giant

ENGLISH

Neptune is very far
away from the Sun

SPANISH

Earth is the planet
where we live on



THANKS!

Do you have any questions?

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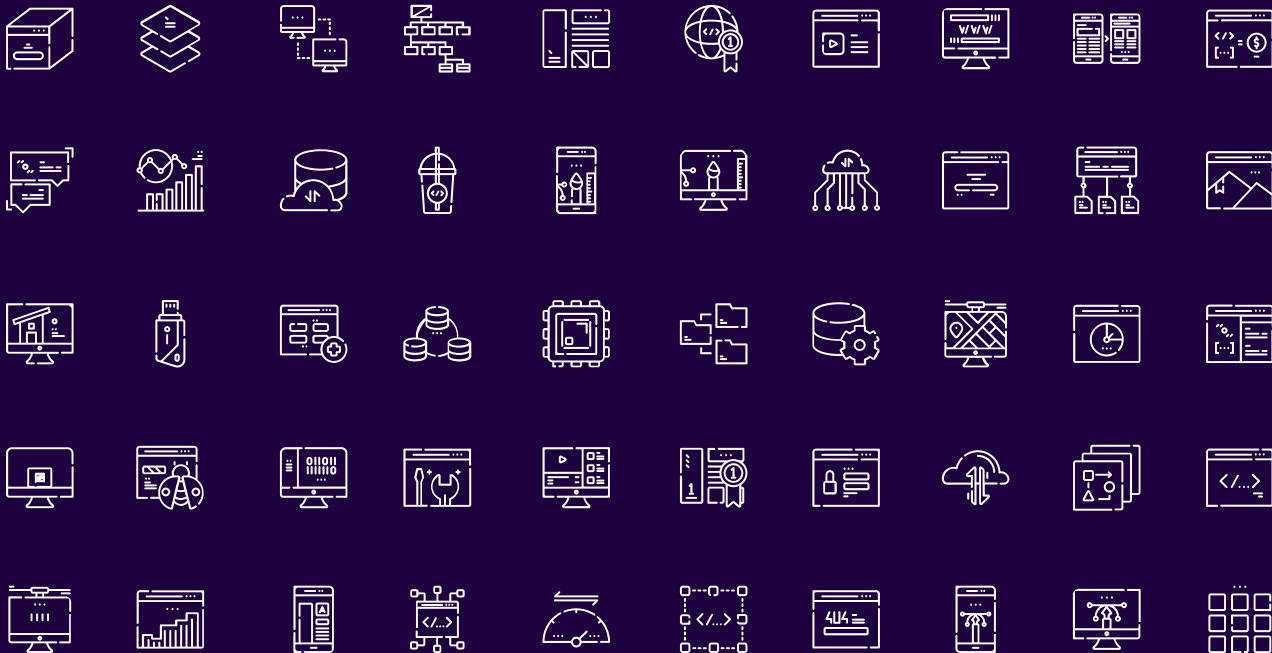
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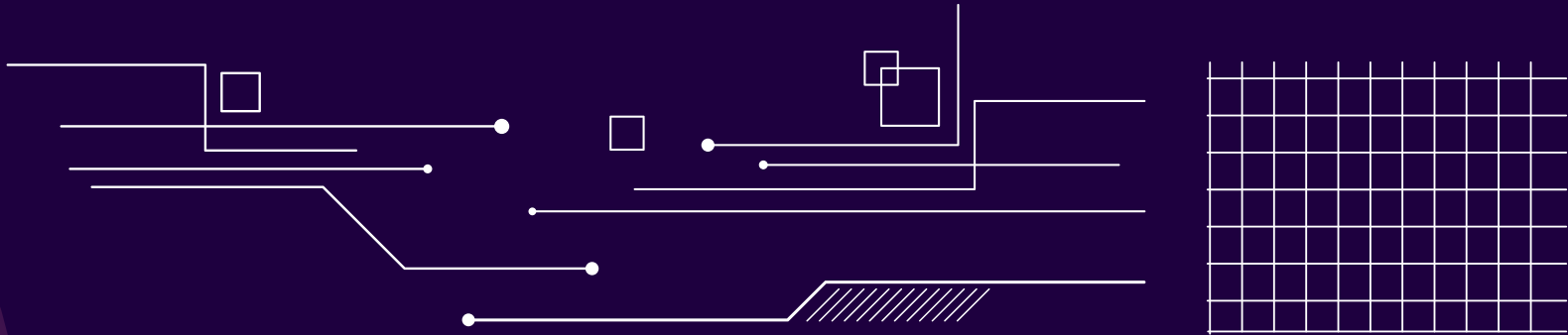


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#ffffff

A pink rounded square with the hex code #ff5e8e in the center.

#ff5e8e

A dark purple rounded square with the hex code #1e003f in the center.

#1e003f

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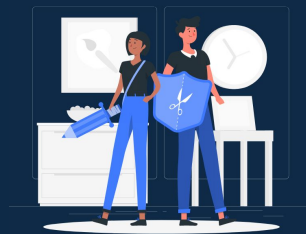
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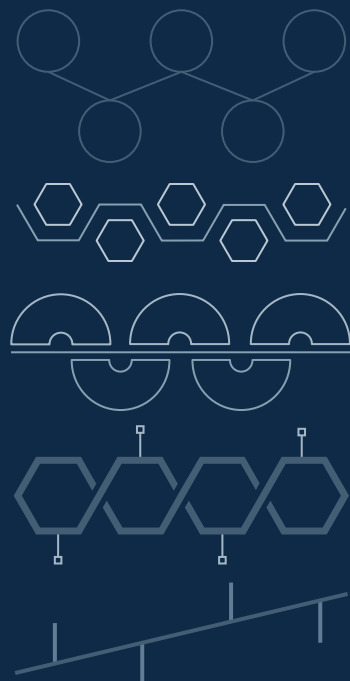
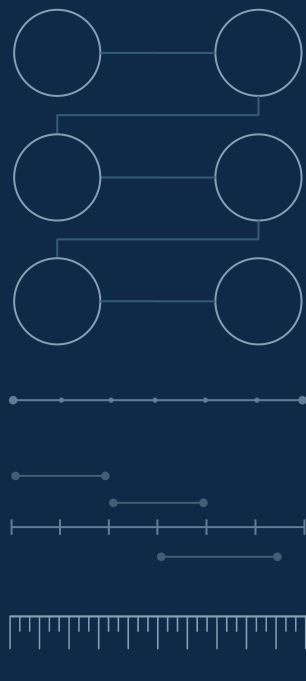
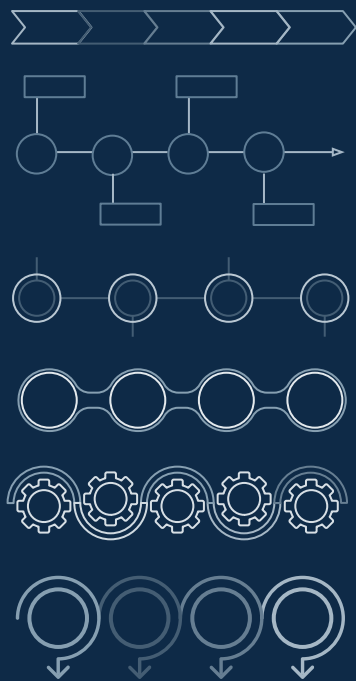
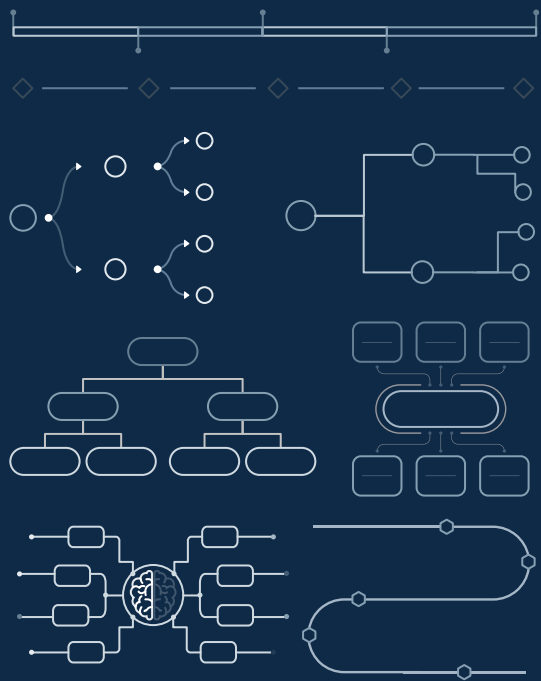
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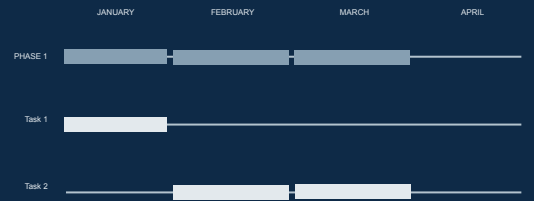
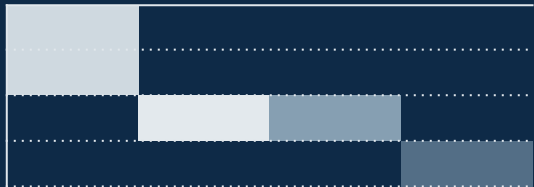
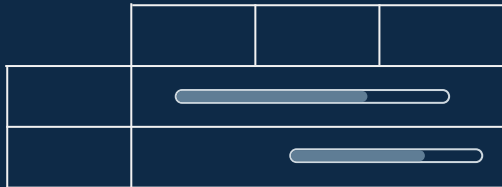
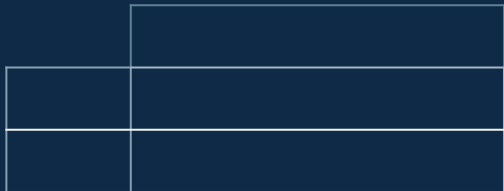
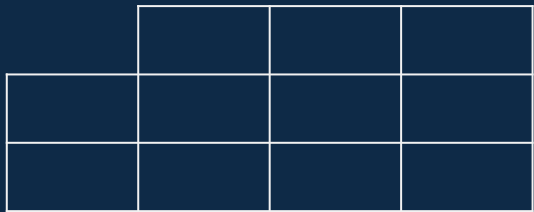
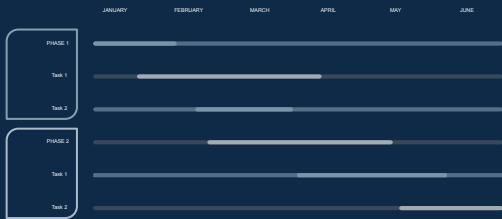
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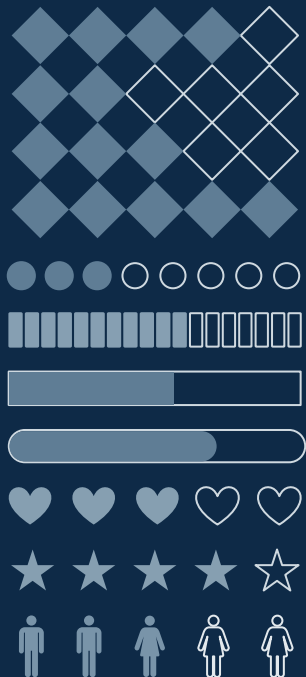
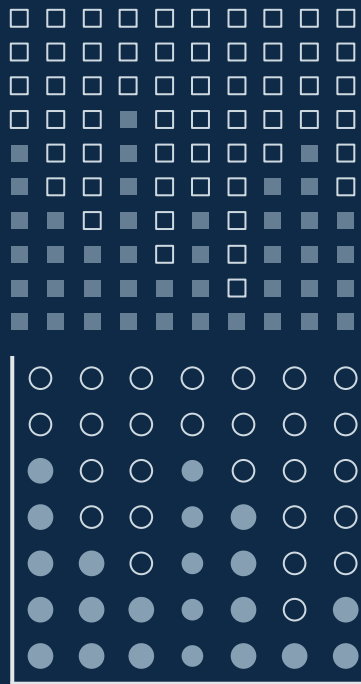












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